

CLOUDSHIELD AI

Master System Specification

Document 00 | Product-Level System Definition | Version 1.0

Field	Value
Document ID	CSA-00-MSS
System	CloudShield AI
Type	Master System Specification
Version	1.0
Status	Baseline / Implementation Precursor
Scope	Entire CloudShield AI product and all major subsystems
DAIL Position	Subsystem of CloudShield AI; not the product itself

Purpose: establish the authoritative product boundary, system model, capabilities, trust model, lifecycle, and engineering structure before implementation.

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1. Executive System Definition

CloudShield AI is a cloud security and infrastructure intelligence platform designed to discover cloud resources, understand configuration and relationships, identify security and operational risk, use AI to explain and propose remediation, and apply a controlled verification-and-promotion lifecycle before a change becomes trusted state.

The product is composed of cooperating layers: product experience and workflows, platform services, AI intelligence, security controls, infrastructure integrations, and DAIL. DAIL (Deterministic AI Infrastructure Lifecycle) is a specialized subsystem that governs candidate infrastructure changes through identity, dependency, impact, verification, evidence, and promotion.

DAIL therefore does not define CloudShield AI's complete scope. It is one mechanism by which CloudShield AI turns AI-generated or human-authored infrastructure changes into controlled, testable, auditable outcomes.

2. Problem Statement & Product Vision

2.1 Problem

Cloud infrastructure is dynamic, interconnected, and security-sensitive. AI-assisted operations can accelerate remediation while also introducing uncertainty about context, side effects, and correctness. CloudShield AI connects discovery, posture analysis, risk reasoning, remediation, deterministic validation, and auditability into one control loop.

2.2 Vision

Provide a continuous control loop: understand the deployed environment, identify what matters, explain why it matters, propose a safe change, deterministically validate it, preserve evidence, and promote only outcomes that satisfy defined trust conditions.

2.3 Product Thesis

The differentiator is not merely adding an LLM to cloud operations. It is combining contextual AI assistance with explicit state, deterministic infrastructure controls, dependency-aware impact analysis, verification, promotion, and evidence.

3. Goals, Non-Goals & Success Criteria

3.1 Goals

- Unified view of cloud assets, configuration, findings, risk, dependencies, and remediation.
- AI for contextual explanation, prioritization, recommendation, and candidate generation, with deterministic safety controls.
- Explicit infrastructure state supporting identity, dependency, impact, verification, and promotion.
- Observable, reproducible, auditable workflows.
- An MVP demonstrating a complete cloud-security-to-remediation loop.

3.2 Non-Goals

- Replacing every cloud-native security product.
- Treating DAIL as the whole product.
- Universal multi-cloud coverage before the core workflow is reliable.
- Unrestricted autonomous cloud administration.
- Uncontrolled autonomous production remediation in the initial trust boundary.

3.3 Success Criteria

Area	Baseline condition
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Discovery	Authorized cloud account connects and resources become normalized assets.
Security	Focused configuration/security checks produce actionable findings.
AI	AI explains findings and produces bounded, traceable remediation candidates.
Lifecycle	Candidates pass impact analysis, verification, decision, and promotion controls.
Trust	Required invariants or verification failures block promotion.
Evidence	Major actions and decisions create durable evidence/audit records.
UX	Users can understand what changed, why, and whether it was verified.
Operations	Health and failure states are sufficiently observable for debugging.

4. Users, Actors & Operating Context

Actor	Primary interaction
Security / Cloud Engineer	Connect environments, investigate findings, review remediation.
Developer / Platform Engineer	Inspect assets, dependencies, configurations, and proposed changes.
Reviewer / Approver	Evaluate and approve/reject high-impact promotions.
Administrator	Manage organizations, projects, access, cloud connections, and policies.
CloudShield AI	Discover, normalize, analyze, reason, propose, orchestrate, verify, record, report.
LLM Provider	Bounded model inference; never the system of record.
Cloud Provider APIs	Authoritative observations and authorized execution surface.
CI/CD / Automation	Optional candidate submission, workflow triggers, and result consumption.

5. Product Scope & System Boundary

Inside the CloudShield AI boundary:

- User interface and workflows
- API/application services
- Asset and configuration model
- Security and risk analysis
- Dependency intelligence
- AI intelligence and remediation assistance
- DAIL lifecycle controls
- Verification and evidence
- Experimentation/evaluation
- Observability and audit

Integrated external boundaries:

- Cloud provider control planes
- Identity providers
- LLM providers
- Source control and CI/CD
- Notification/ticketing systems
- External security data sources

6. CloudShield AI Capability Model

Capability	Purpose	Primary outputs
Cloud Onboarding	Establish authorized cloud access.	Account/environment record
Asset Discovery	Enumerate resources.	Normalized inventory
Configuration Intelligence	Capture and compare configuration.	Snapshots/diffs
Security Analysis	Detect security/policy issues.	Findings
Risk Engine	Prioritize findings using context.	Risk assessments
Dependency Intelligence	Model relationships and blast radius.	Dependency graph
AI Intelligence	Explain and assist reasoning.	Explanations/recommendations
Remediation	Generate bounded changes.	Candidates
DAIL Lifecycle	Control identity, impact, verification, promotion.	Lifecycle decisions/trusted state
Evidence & Audit	Preserve proof and decisions.	Evidence/audit records
Experimentation	Evaluate strategies safely.	Trials/metrics
Reporting	Communicate posture and outcomes.	Dashboards/reports

7. End-to-End Product Workflow

1. Establish project/environment and authorized cloud connection.
2. Discover resources and create a normalized asset inventory.
3. Collect configuration snapshots and associate them with identifiable assets.
4. Run security/policy analysis and generate contextual findings and risk.
5. Build dependency context and potential impact sets.
6. Assemble bounded AI context; explain the issue and/or recommend action.
7. Generate or submit a structured remediation candidate.
8. DAIL validates candidate identity and performs dependency-aware impact analysis.
9. Run deterministic verification gates.
10. Apply human or policy-controlled decision rules.
11. Promote only when required invariants and verification conditions hold.
12. Persist evidence and audit records for actions, results, and decisions.
13. Continuously monitor for drift and re-enter the control loop.

8. System Architecture at Product Level

Authoritative conceptual flow: User Interface → API/Application Layer → CloudShield Core → Product Engines → AI Intelligence / DAIL → Data & Evidence → Cloud Integrations.

Layer	Responsibilities
Experience	Dashboard, asset explorer, findings, risk, remediation review, lifecycle, evidence, reports.
API / Application	Authentication, authorization, orchestration, validation, workflow lifecycle.
Product Core	Projects, environments, accounts, assets, findings, risks, candidates.
Intelligence	Context, security reasoning, recommendations, remediation generation, explanations.
DAIL	Deterministic lifecycle control for infrastructure candidates and trusted state.
Data	Operational state, configuration, evidence/object storage, search/indexing as required.
Integration	Cloud APIs, source control, CI/CD, LLM providers, external systems.
Operations	Logs, metrics, traces, audit, alerting, deployment, recovery.

9. Major Subsystems

Frontend / Product UI

Presents system state and supports investigation, review, approval, and reporting.

API & Application Core

Authoritative product workflow and service boundaries.

Cloud Discovery & Asset Inventory

Connects to cloud APIs and normalizes observed resources.

Configuration & Security Engine

Evaluates configuration and security conditions.

Risk Engine

Prioritizes findings using deterministic and contextual signals.

Dependency Engine

Builds relationships and impact context.

AI Intelligence Layer

Provides bounded contextual reasoning and remediation assistance.

Remediation Engine

Turns findings into structured reviewable candidates.

DAIL Subsystem

Identity, dependency, impact, verification, promotion, and evidence controls.

Evidence & Audit

Makes actions and decisions inspectable and reproducible.

Experimentation

Controlled evaluation of models, prompts, and lifecycle strategies.

Infrastructure & Operations

Secure and reliable runtime foundation.

10. AI Intelligence Model

AI is a reasoning and assistance layer, not the authoritative state store or final safety gate. Model output is converted into structured artifacts and validated before it can affect infrastructure.

AI function	Expected behavior	Control
Context reasoning	Summarize assets, findings, dependencies, history.	Context allowlist/schema
Explanation	Explain why a finding matters using supporting evidence.	Evidence references
Recommendation	Suggest remediation options.	Policy/schema validation
Remediation generation	Produce structured change candidate.	Parsing + validation + DAIL
Natural-language interaction	Answer questions about authorized system state.	Authorization-aware retrieval

Experimentation	Compare prompts/models/strategies.	Evaluation harness
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No raw LLM output may silently mutate trusted infrastructure. Infrastructure-affecting output must enter an explicit candidate and verification lifecycle.

11. DAIL Subsystem Positioning

DAIL begins when a proposed infrastructure state or change requires deterministic lifecycle control and continues through identity, dependency-aware impact analysis, verification, promotion, and evidence.

CloudShield concern	DAIL responsibility
Finding / risk	Consumes contextual inputs; does not own product-wide finding generation.
AI remediation	Consumes structured candidates; does not own all AI reasoning.
Infrastructure identity	Deterministic candidate/resource identity controls.
Dependencies	Lifecycle impact using dependency information.
Verification	Required validation checks.
Promotion	Controlled transition toward trusted state.
Evidence	Lifecycle proof and decision artifacts.
Experimentation	Controlled evaluation of lifecycle strategies.

The existing DAIL specifications remain valid as the detailed subsystem engineering package. They are subordinate to this product-level specification and the future CloudShield AI specifications.

12. Cloud, Data & Trust Model

12.1 Trust hierarchy

- Cloud provider observations are authoritative for observed deployed state.
- CloudShield AI's normalized state is authoritative for its represented model, subject to synchronization.
- AI output is generated/advisory until validated and accepted through deterministic controls.
- Verification results establish whether defined checks passed at a point in time.
- Trusted state is established only through explicit promotion rules.

12.2 Data classes

Class	Examples	Handling
Operational	Users, projects, assets, findings, risks.	Access-controlled database.
Configuration	Snapshots, diffs, proposed configuration.	Versioned/protected.
Sensitive	Credentials, tokens, secrets, private data.	Secure references; minimum AI exposure.
AI artifacts	Context references, outputs, model metadata.	Traceable and retention-controlled.
Evidence	Verification artifacts, approvals, results.	Protected/tamper-evident where feasible.
Audit	Actor, action, time, target, result.	Append-oriented and access-controlled.

13. Core Domain Objects

Object	Definition
Organization	Top-level ownership/tenancy boundary.
Project	Logical workspace for a cloud-security initiative.
Environment	Deployment/context boundary such as dev/staging/prod.
CloudAccount	Authorized provider account/subscription/project connection.
Resource / Asset	Normalized cloud resource representation.
Configuration	Observed or proposed resource configuration.
Finding	Detected security, configuration, policy, or operational issue.
Risk	Contextual impact/priority assessment.
Dependency	Relationship between assets/components.
Candidate	Structured proposed change not yet trusted.
TrustedState	Known accepted state after promotion controls.
Verification	Recorded execution and result of validation gates.

Evidence	Artifact supporting an observation/action/decision.
Experiment	Controlled evaluation of a strategy/model/prompt.
AuditEvent	Record of security/governance-relevant action.

14. State & Lifecycle Model

Conceptual product lifecycle:

DISCOVERED → INVENTORIED → CONFIGURED → ANALYZED → FINDING IDENTIFIED → REMEDIATION PROPOSED → DAIL CANDIDATE → IMPACT ANALYZED → VERIFIED → DECISIONED → PROMOTED → TRUSTED → MONITORED

Not every object uses every state. Detailed subsystem specifications must define exact state machines, valid transitions, guards, failure behavior, idempotency, and evidence requirements.

15. Security & Safety Principles

- Least privilege for cloud and internal permissions.
- Explicit organization/project/environment isolation.
- Secrets are never ordinary application data.
- AI receives minimum necessary context and tool authority.
- No direct autonomous production mutation from raw LLM output.
- Infrastructure candidates are schema-validated and deterministically verified.
- Promotion is policy-controlled and auditable.
- Sensitive evidence has controlled access and retention.
- Failure defaults toward preservation of trusted state.
- Security controls are architectural requirements.

16. Human-in-the-Loop & Governance

Automation level depends on action risk. Informational actions may be automatic; candidate generation may be automatic; sensitive promotion requires verification plus explicit approval or equivalent policy authorization.

Action	Default governance
Discovery/read	Automated within granted permissions.
Analysis/finding	Automated.
AI explanation	Automated with context/evidence controls.
Candidate generation	Automated but non-trusted.
Verification	Automated/deterministic where possible.
Promotion	Policy-gated; human approval for high-risk environments.
Emergency rollback	Explicit operational control with audit.

17. Observability, Evidence & Audit

- Major workflows have correlation/workflow identifiers.
- State transitions record actor, reason, timestamp, source, result, and version identifiers.
- AI-assisted actions preserve relevant model/provider/context metadata subject to security constraints.
- Verification records checks, inputs, outputs, and pass/fail status.
- Promotion references candidate, verification evidence, policy decision, and resulting trusted state.
- Operational telemetry separates metrics, structured logs, traces, and durable evidence.

18. Deployment & Environment Model

Environment	Purpose	Controls
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Local	Developer iteration.	Synthetic/test data; restricted credentials.
Development	Integrated engineering.	Controlled cloud sandbox; CI checks.
Staging	Pre-production validation.	Production-like settings; representative tests.
Production	Real operation.	Strong IAM, monitoring, backups, approvals, audit, change control.

Exact AWS services and Terraform modules belong in infrastructure specifications. This document establishes required behavior and trust boundaries rather than prematurely locking every implementation detail.

19. MVP Definition

- One primary cloud integration, initially AWS.
- Controlled cloud onboarding.
- Resource discovery and normalized asset inventory.
- Configuration snapshots for a focused resource set.
- Focused security checks and actionable findings.
- Risk prioritization with deterministic signals plus AI assistance.
- AI explanation and structured remediation proposal.
- DAIL candidate creation with identity and dependency-aware impact.
- Deterministic verification of bounded remediation candidates.
- Controlled promotion into trusted state.
- Visible evidence/audit trail.
- Dashboard demonstrating the complete workflow.

MVP principle: depth and demonstrable correctness over breadth.

20. Future Scope

- Multi-cloud expansion.
- Broader resource/policy coverage.
- Advanced graph-based risk propagation.
- More capable AI agents with isolated permissions.
- Automated low-risk remediation.
- Advanced drift detection and continuous compliance.
- Source-control and CI/CD integrations.
- Ticketing/incident integrations.
- Enterprise organization policy management.
- Large-scale AI/lifecycle benchmarking.

21. Engineering Documentation Map

ID	Document	Purpose
00	Master System Specification	Product definition and authoritative boundary.
01	CloudShield AI SRS	Complete product requirements.
02	System Architecture	Technical architecture and service boundaries.
03	Functional Specification	Feature behavior and workflows.
04	Product & UX Specification	User journeys and interface behavior.
05	Domain & Data Model	Entities, schemas, relationships.
06	State & Lifecycle	State machines and transition rules.
07	Identity & Dependency	Resource identity and dependency intelligence.
08	Security & Threat Model	Threats, trust boundaries, controls.

09	AI Intelligence Architecture	AI layer architecture and interfaces.
10	LLM Integration	Model/provider integration and structured inference.
11	AI Safety & Guardrails	AI permissions, validation, failure controls.
12–22	DAIL Engineering Package	Detailed DAIL subsystem specifications.
23	AWS & Terraform	Whole-product infrastructure and IaC.
24	API Specification	API contracts.
25	Cloud Integration	Provider integration behavior.
26	External Integration	CI/CD, source control, notifications, etc.
27–30	Operations	Observability, SecOps, DR, deployment.
31–34	Testing & Research	System tests, experiments, AI evaluation, reliability.
35	Master Implementation Roadmap	Implementation sequence and milestones.
36	Definition of Done	Release/engineering acceptance gates.

22. Implementation Readiness & Decision Gates

14. Product boundary approved: CloudShield AI scope and DAIL positioning are agreed.
15. Requirements baseline established.
16. Architecture baseline established: service boundaries, trust boundaries, data flows, deployment model.
17. Domain/lifecycle baseline established: objects, transitions, invariants, ownership.
18. AI safety baseline established: model interfaces, permissions, schemas, validation.
19. DAIL integration baseline established: interfaces with product engines are explicit.
20. Implementation plan approved and traceable to requirements.
21. Build begins with measurable acceptance criteria.

23. Master Acceptance Criteria

- CloudShield AI is clearly defined as the complete product.
- DAIL is explicitly a subsystem with defined responsibilities and interfaces.
- Major product capabilities have architectural ownership.
- Discovery-to-trusted-state workflow is defined.
- AI is bounded by deterministic validation and governance.
- Observed, proposed, verified, and trusted states are distinguished.
- Security, evidence, and audit are first-class concerns.
- MVP demonstrates a coherent end-to-end value path.
- Detailed specifications trace back to this master specification.

24. Appendices & Glossary

Term	Meaning
CloudShield AI	Complete cloud security, infrastructure intelligence, AI assistance, and controlled remediation platform.
DAIL	Deterministic AI Infrastructure Lifecycle; CloudShield AI subsystem for controlled infrastructure-change lifecycle.
Candidate	Proposed infrastructure change/state not yet trusted.
Trusted State	State accepted after required deterministic and governance controls.
Invariant	Condition that must remain true for validity.
Impact Analysis	Assessment of resources/behaviors potentially affected by a candidate.
Verification	Execution of defined validation checks.
Promotion	Controlled transition of an accepted result into trusted state.
Evidence	Information supporting an observation, verification, action, or decision.
Finding	Detected security, configuration, compliance, or operational

	issue.
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Appendix A — Master Architectural Rule

AI proposes. Deterministic systems validate. Governance decides. Cloud APIs execute. Evidence records. Trusted state is earned.